Practical No. 11

Find the Number of CPU Cores and the CPU Manufacturer

1. AIM

To find the number of CPU cores (processing units) and the CPU manufacturer using Unix/Linux commands in a script.

The aim on your sheet says 'C/C++ program', but the given solution is a Bash shell script. Use the shell script, and if the examiner asks, say the same information is read from Linux system commands and files.

2. REQUIREMENTS

- Operating System: Ubuntu / Linux
- Shell: Bash
- Text editor: Nano / VS Code
- Terminal: Linux terminal
- Hardware: computer, keyboard, monitor

3. SHORT THEORY

- CPU (Central Processing Unit): the main component that executes program instructions.
- CPU core: an individual processing unit inside the CPU that can execute instructions. More cores means more tasks can run at the same time.
- CPU manufacturer: the company that makes the processor, e.g. Intel (shown as GenuineIntel) or AMD (shown as AuthenticAMD).
- nproc: shows the number of processing units available to the current process. With hyper-threading this can be more than the physical cores (physical cores x threads per core).
- lscpu: shows detailed CPU information (vendor, model, cores, threads, speed, cache).
- /proc/cpuinfo: a Linux system file that holds processor information. It is a virtual file created by the kernel; there is one block of information per processing unit.

4. ALGORITHM

- 1. Start the script.
- 2. Read the number of processing units using nproc.
- 3. Read the CPU manufacturer from /proc/cpuinfo.
- 4. Display the processing units, manufacturer and model.
- 5. Stop.

5. SHELL SCRIPT (file name: cpu_info.sh)

```
#!/bin/bash
echo "===== CPU INFORMATION ====="
echo "Number of available CPU processing units:"
nproc
echo ""
echo "CPU Manufacturer:"
grep -m 1 "vendor_id" /proc/cpuinfo | awk '{print $3}'
echo ""
echo "CPU Model:"
```

```
grep -m 1 "model name" /proc/cpuinfo | cut -d ':' -f 2
```

6. HOW TO EXECUTE (step by step)

Command / Key	What it does
`Ctrl + Alt + T`	Open the Ubuntu terminal.
`cd ~`	Go to the home folder.
`nano cpu_info.sh`	Create the file, paste the script, save (Ctrl+O, Enter, Ctrl+X).
`chmod +x cpu_info.sh`	Give execute permission.
`./cpu_info.sh`	Run the script. It prints the CPU details.

7. LINE-BY-LINE EXPLANATION

Code	Meaning
`#!/bin/bash`	Shebang. Run this script using the Bash shell.
`echo "===== CPU INFORMATION ====="`	Prints the heading.
`echo "Number of available CPU ... :"`	Prints a label for the next output.
`nproc`	Prints the number of available processing units (for example 8).
`echo ""`	Prints a blank line for spacing.
`grep -m 1 "vendor_id" /proc/cpuinfo`	grep searches for the word vendor_id inside the file /proc/cpuinfo. -m 1 stops after the first match, because the same line is repeated for every core. The line looks like: vendor_id : GenuineIntel
`| awk '{print $3}'`	The pipe | sends the line to awk. awk splits the line into words: $1 = vendor_id, $2 = :, $3 = GenuineIntel. print $3 shows only the manufacturer.
`grep -m 1 "model name" /proc/cpuinfo`	Finds the first line with the CPU model name, e.g. model name : Intel(R) Core(TM) i5-1235U
`| cut -d ':' -f 2`	cut splits the line at the colon. -d ':' sets the delimiter, -f 2 selects the second part, which is the model name itself (it starts with a space).

8. MEANING OF SYMBOLS AND OPTIONS

Symbol	Use
`| (pipe)`	Sends the output of one command as the input of the next command.
`-m 1`	grep option: stop after 1 match.
`-d ':'`	cut option: use : as the delimiter (separator).
`-f 2`	cut option: pick field number 2.
`$3`	Inside awk: the third word of the line. (Not a shell variable.)
`{ }`	Inside awk: the action to perform.
`' '`	Single quotes keep awk's { } and $ away from the shell.
`/proc/cpuinfo`	Path of the system file. / is the folder separator.

Symbol	Use
#	Comment line; #! on line 1 is the shebang.

9. SAMPLE OUTPUT

```
===== CPU INFORMATION =====
Number of available CPU processing units:
8

CPU Manufacturer:
GenuineIntel

CPU Model:
 Intel(R) Core(TM) i5-1235U
```

Your output will depend on your own computer. nproc shows processing units (threads), which may not equal the number of physical cores. Take your screenshot from your own machine.

10. EXTRA COMMANDS (useful in the viva)

Command	Meaning
`lscpu`	Full CPU details in one table.
`lscpu | grep "Vendor ID"`	Shows only the manufacturer. Use this if vendor_id is missing.
`lscpu | grep "Core(s) per socket"`	Physical cores per CPU socket.
`lscpu | grep "Thread(s) per core"`	Threads per core (2 means hyper-threading).
`grep -c ^processor /proc/cpuinfo`	Counts processing units from the file. Gives the same number as nproc.
`cat /proc/cpuinfo`	Shows the whole file.

If the vendor line is empty on your computer (for example on ARM processors like Raspberry Pi), the file has no vendor_id line. Use lscpu instead.

11. CONCLUSION

Thus, the shell script successfully displays the number of available CPU processing units, the CPU manufacturer and the CPU model using the Unix/Linux commands nproc, grep, awk and cut, reading data from /proc/cpuinfo.

12. QUICK VIVA ANSWERS

- CPU: Central Processing Unit. It executes program instructions.
- CPU core: an individual processing unit within a CPU.
- Command for CPU information: lscpu.
- Purpose of nproc: shows the number of processing units available to the current process.
- /proc/cpuinfo: a Linux system file containing processor information.
- Two manufacturers: Intel and AMD.
- Why not exactly physical cores: nproc counts threads too. Cores x threads per core = processing units.